



Technical Description

Select IO availability data missing in 3BSE094064 rev A

Products Concerned

Select I/O

Description

User manual 3BSE094064 rev A does not include availability calculations, these will be added in future revisions of the manual. This document serves as a complement to 3BSE094064 rev A providing availability calculations.

3. Availability Calculations

The total system failure rate can be calculated as the sum of the individual subsystem failure rates.

$$\lambda = \lambda_1 + \lambda_2 + \lambda_3 \dots \lambda_i$$

It is possible, if required, to adapt the calculation of reliability data of I/O modules to the special requirements of a particular application. This can be done in a standardized manner giving good approximated data. The calculation is based on the total failure rate, and certain rules of thumb.



All tables in this document are based on MTTR = 8h unless otherwise indicated.

Functional failure rate: failures that will result in loss of the specified function.

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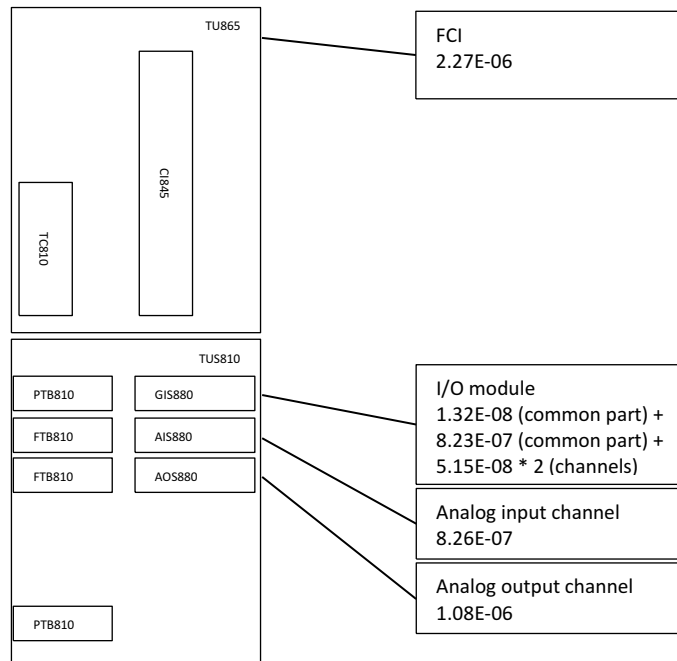


3.1 Single Select I/O system

Function	Including	Fault scenario	Functional failure rate
FCI	1 x CI845 1 x TU865 1 x TC810 1 x HI880	Loss of all channels in the system	2.27E-06
I/O module	1 x GIS880 1 x TUS810 2 x PTB810	Loss of all channels in the system (Common part)	1.32E-08
		Loss of all channels on this TUS810 (Common part)	8.23E-07
		Loss of one channel on this TUS810 (Multiplied for each channel used)	5.15E-08
Analog input channel	1 x AIS880 1 x FTB810	Loss of this channel	8.26E-07
Analog output channel	1 x AOS880 1 x FTB810	Loss of this channel	1.08E-06
Digital input channel	1 x DIS880 1 x FTB810	Loss of this channel	7.63E-07
Digital output channel	1 x DOS880/ DOS885 1 x FTB810	Loss of this channel	1.02E-06

Note: Functional failure rates for xxS810 are identical to xxS880. Failure rate data is calculated at an ambient temperature of 35°C.

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Example 1: Functional failure rate of a single system consisting of FCI, I/O module, one Analog input channel and one Analog output channel:

$$2.27E-06 + 1.32E-08 + 8.23E-07 + 5.15E-08 * 2 + 8.26E-07 + 1.08E-06 = 5.12E-06$$

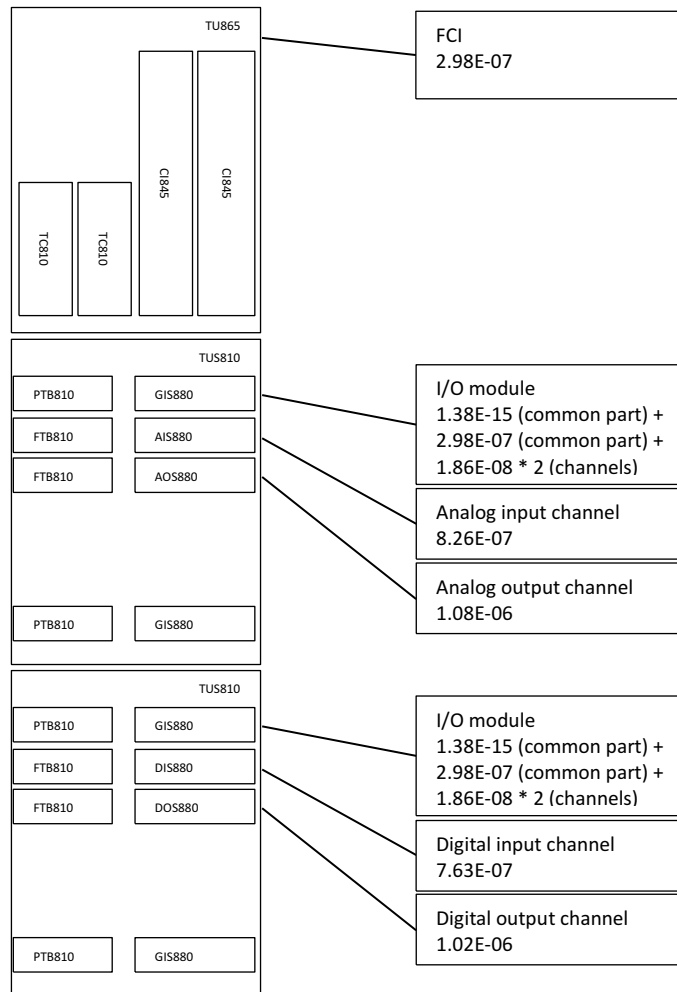


3.2 Redundant Select I/O system

Function	Including	Fault scenario	Functional failure rate
FCI	2 x CI845 1 x TU865 2 x TC810 1 x HI880	Loss of all channels in the system	2.98E-07
I/O module	2 x GIS880 1 x TUS810 2 x PTB810	Loss of all channels in the system	1.38E-15
		Loss of all channels on this TUS810	2.98E-07
		Loss of one channel on this TUS810	1.86E-08
Analog input channel	1 x AIS880 1 x FTB810	Loss of this channel	8.26E-07
Analog output channel	1 x AOS880 1 x FTB810	Loss of this channel	1.08E-06
Digital input channel	1 x DIS880 1 x FTB810	Loss of this channel	7.63E-07
Digital output channel	1 x DOS880/ DOS885 1 x FTB810	Loss of this channel	1.02E-06

Note: Functional failure rates for xxS810 are identical to xxS880. Failure rate data is calculated at an ambient temperature of 35°C.

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Example 2: Functional failure rate of a redundant system consisting of FCI, two separate I/O modules, one Analog input channel and one Analog output channel seated in one TUS810 and one Digital input channel and one Digital output channel in the other TUS810:

$$2.98E-07 + 1.38E-15 + 2.98E-07 + 1.86E-08 * 2 + 8.26E-07 + 1.08E-06 + 1.38E-15 + 2.98E-07 + 1.86E-08 * 2 + 7.63E-07 + 1.02E-06 = 4.66E-06$$

**REVISION**

Rev.ind.	Page (P) Chapt. (C)	Description	Date Dept.
A	All	New document	2020-01-13 IAPCP/REH

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